

Rosefield Solar Project

QGIS Workflow

From installation to buildable area and capacity reconciliation

Gerhardsville Extension 4

Portions 16 and 66 of the Farm Knopjeslaagte 385-JR
City of Tshwane Metropolitan Municipality, Gauteng

Prepared for	Afrosoltech (Pty) Ltd
Project options	20 MWp and 50 MWp on a single site
Coordinate system	Hartebeesthoek94 / Lo29 (EPSG:2053)
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1 Purpose and Scope

This document sets out a complete QGIS workflow for the Rosefield Solar Project, starting from a clean installation and ending with a validated net buildable area and a capacity reconciliation for the 20 MWp and 50 MWp options.

It assumes no prior QGIS experience. Each step states what to click, what to type, and what result confirms the step worked.

1.1 What this workflow produces

- A georeferenced project in the correct South African survey coordinate system
- The site boundary for Portions 16 and 66, validated against published figures
- Constraint layers for servitudes, flood lines, buffers and geotechnical zones
- A net buildable area polygon with a constraint-by-constraint breakdown
- Slope and aspect analysis classified for photovoltaic suitability
- A capacity reconciliation answering the 20 MWp versus 50 MWp question

1.2 Source data

File	Content
GERARDSVILLE X4 - 23 JULY 2025.dwg	Survey drawing: boundary, contours, servitudes, geotechnical zones
GERARDSVILLE EXTENSION 4 - TOWNSHIP.pdf	Township plan, Draft 3, Rev 0, dated 2 August 2024
SERVITUDE REPORT 04082024.pdf	Servitude schedule and SG diagrams
rosefield_site_boundary.geojson	Pre-traced site boundary, accurate to approximately ± 10 m

Status of the source data. The township plan is marked Draft 3, Revision 0, and the flood line, contour, access arrangement and environmental sensitivity certificates are all unsigned. The township is not yet proclaimed. The contour interval is not stated on the plan. Treat all outputs as preliminary until these are confirmed.

2 Software Installation

2.1 ODA File Converter (install this first)

QGIS cannot read DWG files reliably on its own. It calls the ODA File Converter, which must be installed before QGIS starts, or QGIS will not detect it.

1. Download from opendesign.com/guestfiles/oda_file_converter
2. Install with default settings
3. Launch it once to confirm it opens

2.2 Converting the DWG

1. Set **Input Folder** to the directory holding the DWG
2. Set **Output Folder** to the same directory
3. Set **Output version** to **ACAD2018**
4. Set **Output file type** to **DXF**
5. Leave **Recurse Folders** unticked, tick **Audit each file**

6. Click **Start**

Confirms it worked: a `.dxf` file appears beside the DWG, roughly comparable in size. An output file of only a few kilobytes means the conversion failed.

2.3 QGIS

Use the **Long Term Release**, not the latest release. The LTR is more stable and all processing algorithms referenced here are present in it. During installation accept the default components, which include GDAL, GRASS and SAGA.

3 Project Setup

3.1 Create the project folder

Create a dedicated folder before doing anything else, so that relative paths resolve correctly:

```
Rosefield/
  01_source/      DWG, DXF, PDFs, servitude report
  02_gis/         GeoPackage, QGIS project file
  03_output/      maps, exports, reports
```

3.2 Set the coordinate reference system

Open QGIS, then **Project** → **New**. Go to **Project** → **Properties** → **CRS**, type 2053 into the filter, and select **Hartebeesthoek94 / Lo29**.

The Lo29 axis convention. South African Gauss Conformal systems run Y positive **west** and X positive **south**. This is the opposite of the easting and northing convention used almost everywhere else. Coordinates in the QGIS status bar will therefore not look the way you expect. This is correct behaviour, not an error.

If the convention causes difficulty during analysis, **EPSG:32735 (WGS 84 / UTM zone 35S)** is a clean metric alternative covering Centurion, and areas computed in it are effectively identical. Staying in Lo29 is preferred because every other document on this project uses it.

Save immediately as `02_gis/rosefield.qgz`.

3.3 Add satellite imagery

In the **Browser** panel, right-click **XYZ Tiles** and choose **New Connection**.

Field	Value
Name	Esri World Imagery
URL	<code>https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer/tile/{z}/{y}/{x}</code>
Max. Zoom Level	19

Double-click the new entry to add it to the map.

Use Esri World Imagery rather than a Google tile URL. Google's tile service is not licensed for use in third-party GIS applications, and any output built on it would be unusable in a document issued to a client or a funder.

3.4 Navigate to the site

In the coordinate box at the bottom of the window, type the site centroid and press Enter:

```
28.028097,-25.860622
```

Set the scale to roughly 1:10000. The site should read as a long trapezoid of farmland between the R511 and Main Road M26, just south-west of Centurion.

Coordinate order. The QGIS locator bar expects **longitude, latitude** when the project is in a projected CRS. Elsewhere in this document coordinates are quoted as latitude, longitude in line with survey practice. Keep the minus sign on the latitude: without it the location falls in Ethiopia.

4 Load the Reference Boundary

Load the pre-traced boundary **before** importing the DWG. It serves as an independent check that the project setup is correct, and as a reference against which to validate the DWG import.

1. **Layer** → **Add Layer** → **Add Vector Layer**
2. Source type **File**, browse to `rosefield_site_boundary.geojson`
3. Click **Add**, then **Close**

The file is in WGS84 and QGIS reprojects it on the fly.

Style it so the imagery remains visible. Right-click the layer, **Properties** → **Symbology**, select **Simple Fill**, set **Fill style** to **No Brush**, **Stroke color** to a bright yellow, and **Stroke width** to 0.6 mm.

Confirms it worked: the polygon sits over farmland, with the M26 running along its eastern edge and Mimosa Avenue at the north-eastern corner. If it lands in the sea or in another country, the CRS is wrong.

5 Importing the Survey Drawing

5.1 Use the DWG/DXF importer

Do not use **Add Vector Layer** for the DXF. Use the dedicated importer, which preserves CAD layer names. Those names are how you locate the contours, servitudes and geotechnical zones as separate layers.

1. **Project** → **Import/Export** → **Import Layers from DWG/DXF**
2. **Source drawing:** browse to the converted `.dxf`
3. **Target package:** create `02_gis/rosefield_cad.gpkg`
4. Tick **Expand block references**
5. Tick **Use curves**
6. Click **Import**, wait for the layer list to populate
7. Select the layers you need, set **Group name** to `CAD`, click **OK**

5.2 Layers to look for

Content	Likely CAD layer name fragments
Township boundary	BOUNDARY, TOWNSHIP, ERF, CADASTRAL
Contours	CONTOUR, CONT, LEVELS, SPOT
Servitudes	SERV, SERVITUDE, ROW, EskOM, PIPELINE
Geotechnical zones	GEO, DOLOMITE, ZONE, GEOLOGICAL
Flood lines	FLOOD, 1:50, 1:100, FL
Environmental	ENV, SENSITIV, BUFFER

Check this early. Whether the geotechnical zones and contours arrive as proper attributed layers determines roughly two days of effort. If they are present, the constraint build is quick. If they are not, the zones must be digitised by hand from the georeferenced township plan.

5.3 Check the sign convention

South African survey drawings are commonly drawn with negated coordinates so that north appears up in CAD, since Lo29 runs Y west and X south.

Hover the cursor over a known point and read the coordinate display at the bottom of the window.

Observation	Action
CAD layers sit on the GeoJSON boundary	Nothing to do; the import handled it
Layers mirrored or in the wrong quadrant	Apply an affine transform (below)
Layers absent from the canvas	Right-click a layer, Zoom to Layer , and re-check the CRS

To correct negated coordinates, use **Processing Toolbox** → **Vector geometry** → **Affine transform** with **Scale X** and **Scale Y** set to -1 as required.

6 Validation

This is the step that determines whether everything downstream can be trusted. Both checks must pass.

6.1 Check 1: area

Open the attribute table of the boundary layer and click the field calculator. Create a virtual field:

```
Output field name: area_ha
Output field type: Decimal number (real)
Expression:       $area / 10000
```

Polygon	Target	Tolerance
Agricultural erven 1 to 15	131.1828 ha	± 0.5 ha
Full township, erven 1 to 18	141.0505 ha	± 0.5 ha

6.2 Check 2: the northern boundary

Use the **Measure Line** tool along the long northern boundary. It must read **1926.05 m**.

Interpreting the result. If the area is correct but the boundary length is out by 1 to 2 percent, the cause is polygon simplification rather than a scale error, because a scale error would throw the area out by roughly twice the linear error. If both are wrong by the same proportion, the georeferencing is wrong and must be fixed before proceeding.

6.3 Reference values

The pre-traced boundary supplied with this workflow was derived from the vector geometry embedded in the township plan PDF, georeferenced against the Lo29 grid ticks. The drawing scale was confirmed as exactly 1:4000 on A1.

Quantity	Traced	Published
Agricultural area	131.28 ha	131.1828 ha
Northern boundary	1904 m	1926.05 m
Positional accuracy	approximately ± 10 m	

7 Building the Constraint Layers

7.1 Servitude buffers

Servitudes import as lines and must be converted to no-build corridors. Use **Vector** → **Geoprocessing Tools** → **Buffer** with half the servitude width plus a construction tolerance of 2 to 3 m.

Servitude	Width	Buffer	Reference
Electrical power transmission	22.00 m	13.0 m	SG 2337/2000
Right of way (two instances)	15.74 m	10.0 m	SG A5592/1952
Right of way and pipeline	15.00 m	9.5 m	SG 1595/2022
Proposed right of way	15.00 m	9.5 m	Township plan
Proposed right of way	10.00 m	7.0 m	Township plan

Two servitude matters to resolve before finalising.

First, Note 3 on the Extension 4 plan cites a 22.0 m electrical power transmission servitude under SG 2337/2000, while the servitude report attributes that Eskom servitude to Extension 3 on the farm Josudan 801-JR and states that Portion 66 has no servitude traversing it. These statements conflict. Confirm before treating the corridor as a constraint, and note that an Eskom line on or near the site is also a candidate point of connection.

Second, SG 1595/2022 describes a servitude over the **Remainder of Portion 64**, a neighbouring property, which Portion 16 is entitled to use. It lies outside the site and should not be deducted from buildable area.

7.2 Polygon constraints

Flood lines, buffers, environmental sensitivities and geotechnical zones should import as polygons or as closed lines that can be polygonised using **Processing Toolbox** → **Vector geometry** → **Polygonize**.

- 1:50 year flood line
- 1:100 year flood line
- 32.0 m buffer
- Environmental sensitivity areas
- Streets, erven 17 and 18

7.3 The geotechnical layer

This is the layer that carries the most value on this site, because the dolomite classification governs the mounting system, the foundation design and the stormwater strategy.

Add a text field `hazard_class` and a text field `ihc` to the zone layer, then populate per the township plan legend:

Zone	IHC	Hazard class, SANS 1936 (2012)
A	—	Seasonal perched water table, C1/C2-S1/S2
B	2	Dolomite IHC2-D2, seasonal perched water table, C1/C2-S1/S2-H1/H2
C	3/4/5	Dolomite IHC3/4/5-D3, seasonal perched water table, C1/C2-S1/S2
D	3/4/5	Dolomite IHC3/4/5-D3, seasonal perched water table, C1/C2-S1/S2-H1/H2
E	6	Dolomite IHC6-D3/D4, C2-S2
F	7	Dolomite IHC7-D4, C2-S2
G	8	Dolomite IHC8-D4, C2-S2
H	—	Not investigated (agricultural)

Treatment for layout purposes. Zones F and G (IHC7 and IHC8) should be excluded from the array area. Zone E (IHC6) should be treated as conditional pending the dolomite stability investigation report. Zone H is uninvestigated and cannot be relied upon as developable. Driven pile foundations, the default and cheapest mounting solution for utility-scale photovoltaics, are largely unavailable on dolomitic ground because pile penetration creates preferential water pathways and sinkhole initiation risk. Expect ballasted or spread footing systems, or grouted screw piles, with a cost uplift on the mounting package. Stormwater and cable trench design become primary design drivers rather than routine items.

8 Net Buildable Area

8.1 Merge and dissolve

1. **Vector** → **Data Management Tools** → **Merge Vector Layers**, selecting every constraint polygon
2. **Vector** → **Geoprocessing Tools** → **Dissolve** on the merged layer, with no dissolve field

Dissolve before differencing. Constraints overlap. A flood line frequently runs through a servitude corridor, and geotechnical zones cross both. Without dissolving first, overlapping areas are counted more than once and the resulting buildable area is understated.

8.2 Subtract from the site

Use **Vector** → **Geoprocessing Tools** → **Difference**:

```
Input layer:    site boundary
Overlay layer:  dissolved constraints
Output:         net_buildable
```

Read the area with `$area / 10000`.

8.3 Constraint breakdown

For the report, compute what each constraint costs individually. Run **Difference** once per constraint against the site boundary and record the area removed. The sum will exceed the total deduction because of overlaps, and that difference is itself worth reporting.

Constraint	Area removed (ha)	% of gross
Servitudes		
Flood lines, 1:50 and 1:100		
32 m buffer		
Environmental sensitivities		
Geotechnical zones F and G		
Geotechnical zone E (conditional)		
Streets and Erf 16		
Sum of individual deductions		
Dissolved deduction (overlap removed)		
Net buildable		

9 Terrain Analysis

9.1 Build the elevation model

1. Confirm the contour layer carries an elevation attribute. If elevations are absent, the contours are unusable and public 30 m elevation data must be substituted, with the limitation recorded in the report.
2. **Processing Toolbox** → **Interpolation** → **TIN Interpolation**
3. Vector layer: contours. Interpolation attribute: the elevation field. Set **Extent** to the site boundary and **Pixel size** to 2 m.

Contour interval. The township plan prints “CONTOUR INTERVAL: ?? METER” with the value left blank. Confirm it from the DWG, or request it, before relying on the terrain model.

9.2 Slope and classification

Run **Raster** → **Analysis** → **Slope** on the interpolated raster, with **Slope expressed as percent** ticked.

Class	Slope	Suitability
1	< 5%	Preferred; standard mounting and minimal earthworks
2	5 to 10%	Workable; increased earthworks and row adjustment
3	> 10%	Difficult; generally excluded at concept stage

North-facing slopes are advantageous in the southern hemisphere. Run **Raster** → **Analysis** → **Aspect** and note where the site favours north.

10 Capacity Reconciliation

10.1 Design parameters

Derived for the site latitude of -25.8642° , using a no-shading criterion between 09:00 and 15:00 on the winter solstice.

Parameter	Value
Module rating	590 Wp
Module dimensions	2278 × 1134 mm
Module area	2.583 m ²
Rack configuration	2 modules portrait
Collector width	4.58 m
Tilt	25°
Azimuth	Due north
Design profile angle	32.6°
Minimum pitch, no shading	7.17 m
Recommended design pitch	9.5 to 10.0 m
Ground coverage ratio	0.46 to 0.48
Land intensity	approximately 1.3 ha/MWp

Why design wider than the geometry requires. The shading geometry permits 7.17 m, but a design pitch of 9.5 to 10.0 m is preferred for three reasons: margin against diffuse and early or late shading; vehicle access between rows for operations and maintenance; and, specific to this site, wider spacing spreads stormwater runoff rather than concentrating it, which matters on dolomitic ground.

10.2 Capacity formula

$$P_{MWp} = \frac{A_{ha} \times 10^4 \times GCR}{A_{mod}} \times \frac{W_{mod}}{10^6}$$

where A_{ha} is net buildable area in hectares, GCR is the ground coverage ratio, A_{mod} is module area in square metres and W_{mod} is module rating in watts.

As a QGIS field calculator expression on the net buildable polygon:

```
("area_ha" * 10000 * 0.46 / 2.583) * 590 / 1000000
```

10.3 Expected result

Option	Land required	Modules
20 MWp	approximately 26 ha	33 898
50 MWp	approximately 65 ha	84 746

Against an anticipated net buildable area of 85 to 110 ha, both options should be achievable on fixed tilt. Single axis tracking, at 2.0 to 2.5 ha/MWp, would render 50 MWp marginal. The dolomite constraint points toward fixed tilt in any event, so the two findings agree.

11 Outputs

11.1 Map composition

Use **Project** → **New Print Layout** and produce an A3 landscape sheet per theme:

1. Locality and cadastral context
2. Servitudes and rights of way
3. Flood lines, buffers and environmental sensitivities
4. Geotechnical zones with SANS 1936 classification

5. Slope classification
6. Net buildable area with the constraint breakdown

Each sheet requires a title block, north arrow, scale bar, legend, CRS statement and data source note.

11.2 Export

Format	Purpose
GeoPackage	Master GIS deliverable, all layers with attributes
Shapefile	Compatibility with third-party recipients
KML	Review in Google Earth against satellite imagery
GeoJSON	Import into Virto and other web tools
PDF	Map set for issue

Export via **Layer** → **Save As**, setting the CRS to **EPSG:4326** for KML and GeoJSON and retaining **EPSG:2053** for the GeoPackage.

12 Troubleshooting

Symptom	Cause and remedy
DWG import option greyed out	ODA File Converter not installed, or installed after QGIS started. Install it, then restart QGIS.
Layers import but canvas is empty	Right-click the layer and select Zoom to Layer . If the extent is implausible, the CRS is wrong or coordinates are negated.
Geometry mirrored	Negated survey coordinates. Apply an affine transform with scale -1 on the affected axes.
Area reads as a very large number	Layer CRS is geographic rather than projected. Reproject to EPSG:2053 before computing area.
Buffer produces nothing	Input layer is in a geographic CRS, so the buffer distance is being read as degrees. Reproject first.
Difference returns the whole site	Overlay layer is empty or has invalid geometry. Run Vector → Geometry Tools → Check Validity and repair.
Basemap tiles do not load	Check connectivity and the tile URL. Confirm the maximum zoom level is set to 19.
Contours interpolate to a flat surface	The elevation attribute is missing or is stored as text. Convert the field to a numeric type.

A Reference Coordinates

A.1 Site

Point	Latitude	Longitude
Site centroid	-25.860622	28.028097
Centre of Portion 16	-25.860994	28.021332
Centre of Portion 66	-25.861516	28.031802

A.2 Survey grid ticks

These provide an exact georeference. Adjacent ticks are 1000 m apart by construction.

Lo29 Y	Lo29 X	Latitude	Longitude
98 520	2 861 700	−25.860063	28.017150
98 520	2 862 700	−25.869088	28.017075
96 520	2 861 700	−25.860197	28.037100
96 520	2 862 700	−25.869222	28.037027

A.3 Servitude SG 1595/2022

A 15 m right of way and pipeline servitude of 8 292 m² over the Remainder of Portion 64, closing A–B–C–D–E–F–A. Portion 16 is entitled to it. It lies **outside** the site.

Beacon	Lo29 Y	Lo29 X	Latitude	Longitude
A	98 514.47	2 862 139.81	−25.864033	28.017172
B	98 498.35	2 862 142.37	−25.864057	28.017333
C	98 381.79	2 862 603.39	−25.868226	28.018461
D	98 322.38	2 862 593.68	−25.868142	28.019055
E	98 320.03	2 862 608.50	−25.868276	28.019077
F	98 392.96	2 862 620.42	−25.868379	28.018349

A.4 Land schedule

Land use	Erf numbers	Area (ha)	% of township
Agricultural	1 to 15	131.1828	93.00
Infrastructure works	16	3.1586	2.24
Street	17 and 18	6.7091	4.76
Total	18 erven	141.0505	100.00

Erf 16 is designated for infrastructure works and is a natural location for the inverter station, transformer and point of connection compound.

A.5 Title particulars

Item	Detail
Township	Gerhardsville Extension 4 (proposed)
Farm portions	Portions 16 and 66 of Knopjeslaagte 385-JR
Portion 16	120.3837 ha, held under T77021/1997
Portion 66	24.8982 ha, held under T77021/1997
Municipality	City of Tshwane Metropolitan Municipality
Province	Gauteng
Plan reference	DLC Town Plan, Draft 3, Rev 0, 2 August 2024

B Outstanding Information

The following remain outstanding and constrain the analysis:

1. **Dolomite stability investigation report.** The township plan shows percussion boreholes RL1 to RL14 and test pits TP3 to TP104, so an investigation has been carried out. This report is the single most valuable outstanding document.

2. **Contour interval**, not stated on the plan.
3. **Eskom servitude position**, given the conflict between Note 3 on the plan and the servitude report.
4. **Point of connection** and the relevant City of Tshwane feeder or substation.
5. **Land rights status**, and in whose name they are or will be registered.
6. **Offtake route**: wheeling through the Tshwane network, private power purchase agreement, or merchant sale.
7. **Township proclamation status** and the land use application route for a renewable energy facility on agricultural zoning.

Accuracy statement for any issued output. Geometry derived from the township plan is accurate to approximately ± 10 m and is suitable for concept layout, buildable area assessment and capacity reconciliation. It is not suitable for construction, for detailed design, or for reliance by a funder without survey confirmation. Geometry read directly from the survey drawing should be used in preference wherever available.